

Abu Tahim Habiby

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Summary

Research-oriented Computer Science and Engineering graduate with hands-on experience in machine learning, computer vision, and explainable AI. Experienced in developing CNN-, Transformer-, and ensemble-based models using **PyTorch**, **TensorFlow**, and **scikit-learn** for medical image analysis and clinical prediction. Seeking research-based graduate opportunities in computer vision, deep learning, and applied AI.

Research Interests

Machine Learning, Deep Learning, Computer Vision, Medical Image Analysis, Explainable AI

Education

Bachelor of Science in Computer Science and Engineering 2026

Daffodil International University, Dhaka, Bangladesh

CGPA: 3.81/4.00

Credits: 148

Relevant Coursework: Data Structures & Algorithms, Artificial Intelligence, Statistics, Database Systems, Software Engineering

[Daffodil International University Website](#) 🔗

Research Experience & Publications

ClinFusion-Net: Hybrid CNN–Transformer Model for GI Disease Classification (Undergraduate Thesis) 2026

PyTorch, ConvNeXt-XL, Swin-Large, Explainable AI

- Developed a hybrid CNN–Transformer model for GI disease classification on **12,000** Kvasir images.
- Achieved **95.1%** test accuracy in version 2 and **93.75%** test accuracy in version 1.
- Integrated **Grad-CAM**, **LIME**, and **t-SNE** for model interpretability; live demo available on [Hugging Face Spaces](#) 🔗.

A High-Accuracy Stacking Ensemble Model for Precise Thyroid Disease Detection with SHAP and LIME 2026

Explanations

Python, scikit-learn, SHAP, LIME

- Developed a stacking-based machine learning framework for thyroid disease prediction.
- Combined **Random Forest**, **XGBoost**, and **Decision Tree** with **Logistic Regression**; applied **SHAP** and **LIME** for explainability.
- Published in **Informatics in Medicine Unlocked (Elsevier)**. DOI: [10.1016/j.imu.2026.101764](https://doi.org/10.1016/j.imu.2026.101764) 🔗.

RXV: Interpretable RF-Stacked CNNs for Brain Tumor MRI Classification 2025

TensorFlow, CNN Ensembles, Random Forest

- Built an RF-stacked ensemble using **ResNet50V2**, **VGG16**, and **Xception**.
- Evaluated the model on **6,004** balanced brain MRI images and achieved **99.56%** test accuracy.
- Published in **IEEE WIECON-ECE 2025**. DOI: [10.1109/WIECON-ECE69386.2025.11525916](https://doi.org/10.1109/WIECON-ECE69386.2025.11525916) 🔗.

An Explainable Stacking Ensemble Framework for Early Heart Attack Risk Detection 2026

Python, scikit-learn, SHAP, LIME

- Developed an explainable stacking ensemble on **1,888** patient records with **14** clinical features.
- Achieved **96.83%** accuracy and **96.76%** F1-score using preprocessing, **SMOTE**, stratified **5-fold cross-validation**, **SHAP**, and **LIME**.
- Available on **ResearchGate**. Preprint DOI: [10.13140/RG.2.2.15388.55687](https://doi.org/10.13140/RG.2.2.15388.55687) 🔗.

Selected Technical Projects

AI Research Paper Assistant

2025

Python, Streamlit, Sentence Transformers, FAISS, Groq API, PyPDF

- Built a RAG-based research assistant for grounded question answering, structured summaries, and source-aware exploration of research papers.
- Links: [GitHub](#) | [Live Demo](#)

AI Codebase Assistant

2025

Python, Streamlit, Sentence Transformers, FAISS, GitPython

- Developed a repository-level assistant for codebase understanding, file-level explanation, and developer onboarding support.
- Links: [GitHub](#) | [Live Demo](#)

Academic Hub: Data-Driven Full-Stack Platform

2025

Laravel, MySQL, Google OAuth, Google Drive API

- Built a full-stack academic portal for CGPA tracking, course management, and academic workflow support.
- Integrated **Google OAuth** and **Google Drive API**; demonstrated database design and software engineering experience.
- Link: [GitHub](#)

Skills

- **Programming:** Python, C, C++, Java
- **Databases:** SQL, MySQL
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow/Keras, scikit-learn, Pandas, NumPy, Matplotlib, Transformers
- **Explainability & Retrieval:** Grad-CAM, SHAP, LIME, FAISS, Sentence Transformers
- **Development Tools & Environments:** Git, GitHub, Jupyter Notebook, Google Colab, VS Code, Streamlit, Hugging Face Spaces

Languages

- **Bangla:** Native
- **English:** IELTS Academic 6.5, CEFR B2

Competitions & Awards

- **Top 5 Finalist** – AI-FICATION 2025, National AI Competition, CUET; selected among **80** teams.
- **Finalist** – Crack Dataset Challenge 2024, DIU NLP & ML Lab.
- **Finalist** – Take-Off Programming Contest 2022, DIU Computer Club; ranked among the top **50** of **350+** participants.

Community Involvement & Leadership

- **Press Secretary, Daffodil AI Club** 2025–Present
Led communications and event outreach; supported technical events and community engagement.
- **Volunteer Coordinator, Doorbin Foundation** 2024
Coordinated a free medical camp for **200+** participants and flood relief support for **150+** families; worked with a **50**-volunteer team.

Certifications

- [Supervised Machine Learning: Regression & Classification](#) – Stanford Online, Coursera
- [Advanced Learning Algorithms](#) – Stanford Online, Coursera
- [Machine Learning with Python](#) – IBM, Coursera
- [Deep Learning with Keras & TensorFlow](#) – IBM, Coursera
- [Introduction to Deep Learning & Neural Networks](#) – IBM, Coursera
- [Data Science & Machine Learning with Python](#) – aiQuest Intelligence, 50-hour course
- [Database Training: MySQL & Oracle SQL](#) – ICT Division, Bangladesh EDGE, 80-hour training